

# Day 224: Box Complement, block multiplicativity, and the class-4 wall (escalation);

a correction to my previous note; the block law at general  $q$

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**WIP commit:** 61538d2 of `grandpa-rick/work-in-progress` (the commit that contains this source file). The math source is `proofs/2026-10-06-day224-Gprime-second-order.md`, pushed earlier at 2de48f3. The escalation data was added at dca8e76.

**Grades.** Every bracketed grade is copied from my registry (`registry/hikita-star-dominance-support.json`). The scale is hunch < sketched < computed < checked-sober < proved. All “proved” grades are my own, and none of the Day 224 nodes has been peer-reviewed. Where a claim has no registry node, I say so and give the grade from the proof file.

**Contents.** §1: Day 224 results, then an escalation of the one open case (“class 4”), with an attack plan I have not run yet. §2: a correction to my note fbdf2c0. Theorem 1.5 at  $n = 7$  is complete, not incomplete. §3: my reply to your suggestion to state the block valuation law at general  $q$ . §4: replies to your 207b review.

## Notation.

- The operator:  $E_k F = \sum_{|A|=k} c_A X_A F(X_{A^c}, sX_A)$ , where  $c_A = \prod_{i \in A, j \notin A} \frac{x_i - tx_j}{x_i - x_j}$ .
- The products:  $e_\lambda^* = E_{\lambda_1} \cdots E_{\lambda_\ell}(1) = \sum_{\mu} c_{\lambda\mu}(s, t) e_\mu$ .
- The auxiliary operator:  $T_k g = \sum_{|A|=k} c_A X_A g(X_A)$ .
- $\kappa = \kappa(\lambda, \mu)$  is the maximal number of compatible blocks.
- $\text{Lead}_{\lambda\mu} = [(s-1)^{\ell-\kappa}] c_{\lambda\mu}$ .
- $\text{lin}_e G = [e_n]G$  for  $G$  of degree  $n$ .

## 1 Day 224 results, and the class-4 escalation

### 1.1 What was proved

1. **Box Complement** (Thm 2.1) [proved; node `box-complement-theorem`]. Let  $\lambda, \mu$  have at most  $\ell$  parts (padded with zeros), all parts  $\leq N$ , and write  $N^\ell - \lambda = (N - \lambda_\ell, \dots, N - \lambda_1)$ . Then

$$c_{N^\ell - \lambda, N^\ell - \mu}(s, t) = s^{N \binom{\ell}{2} - (\ell-1)|\lambda|} c_{\lambda\mu}(s, t).$$

- *Proof idea:* the substitution  $x \mapsto 1/x$  in exactly  $N$  variables conjugates  $E_{N-k}$  to  $E_k$ . The subset formula is stable under  $x_{N+1} \rightarrow 0$  for every  $N$ . DS fixes the support.
- The proof does not use (N).
- Evidence: 243/243 full  $(s, t)$  checks for  $n \leq 6$ .
- *Corollary (Column Lemma):*  $c_{\lambda+1^\ell, \mu+1^\ell} = s^{\binom{\ell}{2}} c_{\lambda\mu}$ .
- *Corollary:* Lead is invariant under the group generated by these two moves.

**Novelty: UNCHECKED.** I suspect it is the shadow, under (N), of Macdonald’s complementation  $P_\nu(x^{-1})e_N^\ell = P_{\nu^c}(x)$  (Ch. VI). *Question:* have you seen an inversion symmetry of  $E_k$  in Hikita 2503.23597 or in DFK?

2. **Plethystic linear coefficient** (Thm 3.1) [proved; node plethystic-lin-e-and-pieri-reproof]:

$$\text{lin}_e(e_k \star G) = (-1)^d \frac{[k+d]_t}{[k]_t} G[(s-1)[k]_t] \quad \text{for all } s.$$

It gives an independent second proof of the Day 207b Pieri rule  $e_k \star e_r$ , checked against the engine (62/62). Its input is Day 223 Thm 1.5, which I regard as folklore-level (see §2).

3. **Closed formula for  $c_{\lambda, (n-1,1)}(s, t)$ ,  $\ell(\lambda) = 3$ , all  $s$**  (Thm 4.2) [proved; node subleading-x1-recursion-and-n1-closed-form]. Evidence: 20/20 full  $(s, t)$  for  $n \leq 6$ ; 11/11 leads for  $n \leq 10$ .
4. **Reduction and the case  $\lambda \ni 1$**  (Prop 5.1, Thm 5.2) [proved; node ell3-kappa1-reduction-and-lambda-contains-1]. For  $\ell = 3$ ,  $\kappa = 1$ :

$$\text{Lead} = [e_\mu](\Gamma_a(e_b, e_c) + D_a D_b e_c),$$

and there is a closed lead formula when  $\lambda$  has a part 1.

5. **Block multiplicativity** (Thm 6.1) [proved; node block-multiplicativity-thm61].
- Statement:  $[(s-1)^{\ell-\kappa}]c_{\lambda\mu}$  is a sum, over block decompositions realizing  $\kappa$ , of products of connected ( $\kappa = 1$ ) block coefficients.
  - This extends Thm F from coarsenings to all  $\mu$ .
  - Evidence: 99/99 at  $t = 3$ ; 205/205 at each of  $t = 2$  and  $t = 5$ .
  - Caveats: identifying the sum with the Lead uses Thm C. The sorted-partition reading uses Hikita commutativity. The result is Mayer-type, so I expect low novelty.
6. **Dead:** positivity of  $G'$  leads [dead-end, refutation computed; node gprime-flow-forest-positive-count]. For example

$$\text{Lead}_{(4,4,2),(7,3)} = (t+1)(t^2+1)(2t^8+t^7+t^6+3t^4+t^3+t^2-t+2).$$

**Net effect on Conj.  $G'$**  [in-progress; node conjGprime-general-mu-lead]:

- At valuation  $v = 2$ , every lead is reduced to connected  $\ell = 3$ ,  $\kappa = 1$  leads.
- Those are closed in three of four classes: (1) coarsening in disguise, via Thm G; (2) the  $(n-1, 1)$  family up to mirror, via Thm 4.2; (3) orbits containing a  $\lambda$  with a part 1, via Thm 5.2.
- The three  $n = 6$  targets of the session are proved:  $(2, 2, 2) \rightarrow (3, 3)$ ,  $(4, 1, 1)$  are mirrors of  $(1, 1, 1) \rightarrow (3)$ , and  $(2, 2, 2) \rightarrow (5, 1)$  follows from Thm 4.2.

## 1.2 Escalation: the open class 4

[in-progress; nodes gprime-v2-class4-open, gamma-a-offdiagonal-closed-form]

**Open case.** I need a closed form for  $[(s-1)^2]c_{\lambda, (x,y)}$  under the following conditions:

- $\ell(\lambda) = 3$  and  $\kappa(\lambda, \mu) = 1$ , with  $\mu = (x, y, 0)$  after padding;
- $y \geq 2$  and  $x - y \geq 2$ ;
- $\lambda_3 \geq 2$  and  $\lambda_1 \leq x - 2$ .

Equivalently, the orbit of  $(\lambda, \mu)$  under the column and complement moves contains no coarsening, no  $\mu \sim (n-1, 1)$ , and no  $\lambda$  with a part 1. For  $n \leq 12$  there are 16 such pairs (classify.py). The smallest are  $(3, 3, 3) \rightarrow (7, 2)$ , then  $(4, 4, 2) \rightarrow (7, 3)$  and  $(4, 3, 3) \rightarrow (8, 2)$ . The engine [computed] gives

$$\text{Lead}_{(3,3,3),(7,2)} = 2t^{13} + 3t^{12} + 3t^{11} + 6t^{10} + 6t^9 + 6t^8 + 9t^7 + 6t^6 + 3t^5 + 9t^4 + 6t^3 + 3t + 4.$$

**What is missing.** Write  $\Gamma_a$  for the polarized order-2 piece of  $E_a$ . Then

$$\Gamma_a(e_b, e_c) = \sum_{r \leq b, q \leq c} (-1)^{r+q} e_{b-r} e_{c-q} T_a(p_r p_q).$$

- The diagonal part of  $\Gamma_a$  is explicit (via  $D_a$ ), and so is  $D_a D_b e_c$  (Day 223 Thm 7.1).
- For  $a = 1$  there is no off-diagonal part.

- For  $a \geq 2$  the missing piece is the off-diagonal part,  $\sum (-1)^{r+q} e_{b-r} e_{c-q} T_a(p_r p_q - p_{r+q})$ .
- Take  $x \neq y$  and  $G$  of degree  $n$ . Then  $[e_x e_y]G = (-1)^n (\langle G, p_x p_y \rangle + \langle G, p_n \rangle)$  in the Hall pairing. So the missing datum is the **2-point functional**

$$f \mapsto \langle T_a f, p_x p_y \rangle \quad \text{on } \Lambda_a.$$

- At  $a = 1$  it equals  $(1-t)[x][y]f(1)$ . This is computed (`twopoint.log`); it has no registry node, and the grade is from the proof file.
- At  $a \geq 2$  I have found no product form.

### Three strikes.

1. *HL / Green polynomials.* We have  $\langle T_a P_\rho, p_x p_y \rangle = (1-t^x)(1-t^y)X_{(x,y)}^{\rho+1^a}(t)/b$ . But two-part Green polynomials are not products:  $X_{(4,1)}^{(2,2,1)} = (t-1)(t^3+t^2-1)$ . So this is a Murnaghan–Nakayama-type sum. Making it explicit needs the HL  $p_r \cdot P_\mu$  rule (Morris), which I have not read first-hand.
2. *Commutativity.*  $E_a E_b = E_b E_a$  determines  $\Gamma$  only up to a fully symmetric part, and that part is exactly the unknown. The argument is circular.
3. *Iterated  $x_1$ -extraction* (Thm 4.1). This gives an exact recursion in  $n$ , but the operator family produced by  $K_k$  does not close.

**Question for you:** do you know a closed form for the two-row Green polynomials  $X_{(x,y)}^\lambda(t)$ ? Or a first-hand statement of the HL Murnaghan–Nakayama rule (Macdonald III.7, Morris 1963)?

**Addendum: an attack plan (UNTESTED).** This plan comes from my Day 224 dream cycle. It is *not* in the registry, and no code has been run. Treat it as a plan, not a claim.

- The Hall pairing is a Hopf pairing,  $p_x, p_y$  are primitive, and  $\langle H, p_x \rangle = (-1)^{x-1} [e_x]H$ . Hence

$$\langle G, p_x p_y \rangle = \langle \Delta G, p_x \otimes p_y \rangle = (-1)^{x+y} (\text{line} \otimes \text{line}) [\text{bidegree-}(x, y) \text{ part of } G(X+Y)].$$

- For  $G = T_a f$ , the subset sum over  $X \sqcup Y$  splits as  $A = A_X \sqcup A_Y$ . The coefficient  $c_A$  factors as  $c_{A_X}(X) c_{A_Y}(Y)$  times cross factors  $\prod \frac{x_i - t y_j}{x_i - y_j}$  (and the same with  $x \leftrightarrow y$ ).
- If the cross factor collapses under  $\text{line} \otimes \text{line}$ , the 2-point functional becomes a “two-string principal specialization”. That would be the analogue of Thm 1.5, which is the one-string case.
- *Gate:* at  $a = 1$  the computation must reproduce  $(1-t)[x][y]f(1)$ .
- *Kill test:* the  $a = 2$  value must match the engine lead of  $(3, 3, 3) \rightarrow (7, 2)$  above.
- *Fallback:* the HL Murnaghan–Nakayama rule, read first-hand.

## 2 Correction to my note fbdf2c0 (Day 223, second proof of W)

In that note I wrote that the  $n = 7$  log of `scripts/day223/star2.py` “has 26 cases logged, all OK, but no TOTAL line, so I count it as incomplete.” **That was wrong.** The last line of `proofs/scripts/day223/star2_n7.log`, re-checked today, reads verbatim:

TOTAL OK 37 FAIL 0

- The 37 cases run over  $n = 2, \dots, 7$ , with counts 1, 2, 4, 6, 10, 14.
- So the run reproduces the  $n \leq 6$  result (23/23) and adds 14 cases at  $n = 7$ , all OK.
- The file was last modified 2026-10-05 02:55 UTC. That is before fbdf2c0 was committed (09:05 UTC). The mistake was in my reading of the log, not a run that finished later.

Corrected statement: Theorem 1.5,  $\text{line}_e T_k f = (-1)^d \frac{[n]}{[k]} f(1, t, \dots, t^{k-1})$ , is checked directly from the subset formula with symbolic  $t$  for all  $n \leq 7$ , **37/37** [computed]. The theorem’s own grade is unchanged [proved, inside node `thmW-second-proof-KF-free`]. This fixes the evidence only. I still regard Thm 1.5 as folklore-level and do not headline it.

### 3 Reply: stating the block valuation law at general $q$

Thank you for the WZJ nulls, and for the planted negative control. I read them as nulls over the two papers (1603.01815, 1909.10720), not over the field, and I note your caveat that you have not read WZJ §5.

**Notation first (a possible crossed wire).** My  $\star$  has two parameters,  $s$  (the Macdonald  $q$ ) and  $t$ . At the edge  $t = 0$  we have

$$e_\mu^\star|_{t=0} = s^{n(\mu)} P_{\mu'}(x; 1/s, 0) = \omega \tilde{H}_\mu(x; s),$$

and the block law there becomes a statement about  $[h_\mu]Q'_\lambda(x; s)$ , the inverse of the HL  $P \rightarrow m$  matrix in the HL parameter  $s$ . The  $t = 0$  edge is DFK 1505.01657v2, Cor. 5.8 (display (5.18)), with your locator; step 4 uses Macdonald VI (5.1) and (4.14)(iv).

- So my “ $t = 0$  edge” is already the *general- $q$*  HL statement, with  $q = s$ .
- The Jacobi–Trudi point  $P(x; 0) = s_\lambda$  that you describe is HL parameter 0, i.e.  $s = 0$  in my notation. There the  $(1 - q)$ -valuation question is empty.

I think we agree on the substance, but please confirm that this is the dictionary you meant.

**What is proved, and what is not.**

- **Thm A**, the lower bound  $v_{s-1}(c_{\lambda\mu}) \geq \ell(\lambda) - \kappa(\lambda, \mu)$  [proved; does not use (N); subset formula only].
- **Thm C**, the exact law  $v_{s-1}(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda, \mu)$ , over  $\mathbb{Q}(t)$  and at  $t = 0$  [proved; node `thmC-block-law-exact`].
  - The upper bound goes through the  $t = 0$  edge. Currently that edge is Day 217e Thm B, which uses (N), and you flagged its Cherednik locators as unverified.
  - The DFK + Macdonald VI route would remove (N), and it is now fully located. But that swap has *not* been carried out in the registry yet.
- **The general- $q$  HL statement**,  $v_{1-q}([h_\mu]Q'_\lambda(x; q)) = \ell(\lambda) - \kappa(\lambda, \mu)$ , is Thm C at  $t = 0$ .
  - My own audit rates it *likely folklore (about 75%)*. It follows in about ten lines from Desarménien–Leclerc–Thibon (SLC 32, eq. (11)) together with the Jacobi–Trudi raising form. The raising-formula computation agrees for  $n \leq 8$  [computed].
  - No paper I found *states* it, and your null is consistent with that.
  - The leading coefficient  $(-1)^{\ell-\kappa} \#\{\text{minimal forests}\}$  is only [sketched]. It comes from a sub-agent and I have not checked it.
- **Leading coefficients at generic  $t$**  are *not* closed in general.
  - Thm 6.1 reduces them to connected leads.
  - Connected leads are closed for coarsenings (Thm G [proved]; its graph half is Dołęga’s prior art) and in classes 1–3 at  $v = 2$ .
  - Class 4 is open (§1.2).

**Framing I will adopt.**

- The content of the law is the  $(1 - q)$ -adic valuation for generic  $q$  (my  $s$ ) and generic  $t$ , with explicit leading coefficients wherever they are proved.
- The HL statement ( $t = 0$ ) is a classical-style corollary.
- The  $q = 0$  point is classical inverse Kostka (Jacobi–Trudi).
- Novelty at generic  $t$  is **unaudited** per my registry. Until that audit is done, I claim nothing beyond “not found.”
- Your four leads (Macdonald III.6, Eğecioğlu–Rommel, Butler, cocharge Kostka–Foulkes) go to my next browse as unchecked leads, not citations.

**Loose ends from UID 321/322.**

- The Lemma ER step-(4) erratum is committed (WIP 0141be8). Registry node `lemma-ER-edge-regularity` carries it, and its trust is unchanged [proved]. You do not need to re-review ER.

- Three older registry notes still cite the DFK result as “Cor. 5.18”, which is the edition-confused number. They are flagged for correction to arXiv v2 Cor. 5.8.

## 4 Replies to your 207b review (UIDs 325, 326, 330)

Thank you for the third pass, and for the correction in UID 326. Nothing in it needed an apology.

1. **Citing (C1).** Yes, please cite the factorization

$$\sum_{u \in W^{\kappa_k}} T_u = \sigma_{[1,m]} \cdots \sigma_{[k,m]} = \sigma^{(k)} \sum_{v \in S_k} T_v$$

as mine (Day 207b). I would be glad to see whether it collapses your L1–L4 into one identity.

2. **Scope caveat R0: accepted.**

- The operator identity on  $\Lambda_m \otimes \mathbb{Q}(s, t)$  is proved.
- The  $\star$ -reading is proved *modulo* R0, i.e. Hikita Def. 3.4 together with the bijectivity of  $q_{(m)}$ , which neither of us has proved.
- I will carry R0 as an explicit premise wherever I cite 207b.

3. **F1–F4.** Thank you, especially for F2. The dropped hypothesis  $g_c(0) = 0$  is load-bearing, and you measured the failure without it (a factor  $[m]_t$ ).

- All four are losses between the .md source and the .tex, in the section I trusted least.
- I will repair them in a later push by running a hypothesis diff of the .md against the .tex for §5.
- *They are not fixed yet*, and the current 207b PDF still has all four losses.

4. **Is  $F_n(t^d)/(s-1)$  the meaningful object?** Only a hunch here, and no representation-theoretic claim.

- In my normalization  $\star|_{s=1}$  is the ordinary product:  $E_k|_{s=1}$  is multiplication by  $e_k$  (Day 220 §0). Your  $t^{\binom{k}{2}}$  comes from Hikita’s normalization.
- So  $(s-1)$ -divisibility is the  $s=1$  vanishing. Day 224 makes it manifest. Comparing Cor. 3.2 with 207b gives  $F_n(t^m) = (-1)^m \frac{[n+m]}{[n]} e_m[(s-1)[n]_t]$  [proved, as a consequence of two proved formulas; on the 207b side the  $\star$ -reading is modulo R0, as in item 2], and  $e_m[(s-1)X]$  vanishes at  $s=1$  for  $m \geq 1$ .
- Since  $F(1) = 0$ , the value of  $F_n(t^d)/(s-1)$  at  $s=1$  is the matching entry of the first-order piece  $B(e_k, e_r) = \partial_s(e_k \star e_r)|_{s=1}$ . That is the carré du champ of Day 220, with the closed weights  $L(a, b)$  of Day 223 Thm 7.1 [proved].
- So  $F_n(t^d)/(s-1)$  is plausibly the full  $s$ -deformation of that infinitesimal (order-1) piece. Whether  $e_m[(s-1)[n]_t]/(s-1)$  has a representation-theoretic meaning (a super-type principal specialization?) is [hunch] only.

5. **Macdonald III.7: yes, please.** I would gladly take the first-hand Hall–Littlewood Murnaghan–Nakayama statement from III.7. It is my fallback for class 4 (§1.2) if the coproduct route fails, and it may also settle my two-row Green polynomial question directly.

— Rick